

The **pongtex** Package

Typesetting Riichi Mahjong Hands and Discard Rivers

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Abstract

pongtex typesets Japanese/Riichi Mahjong tiles from compact MPSZ notation. It provides a document command, `\mahjong`, for hands and melds, and `\mahjongriver` for discard rivers. It supports red fives, face-down and blank tiles, rotated and stacked tiles, concealed-kong shorthand, local and global size options, and recolouring of tile backs.

1 Loading the package

A minimal document looks like this.

If you are working from the CTAN source archive, generate the runtime package file first:

```
latex pongtex.ins
```

The distributed user manual lives in `doc/pongtex-doc.pdf`; its source is `doc/pongtex-doc.tex`. The package implementation source is `pongtex.dtx`, and `pongtex.ins` extracts `pongtex.sty`.

```
\documentclass{article}
\usepackage{pongtex}
\begin{document}
\mahjong{111m456s111p11122z}
\end{document}
```

The package expects the PDF artwork in the `tiles/` directory. With the default setting, `pongtex.sty` and `tiles/` should be installed in the same TeX-searchable package directory.

2 Package options

Options can be supplied when loading the package or changed later with `\pongtexsetup`.

```
\usepackage[
  height=1.5\baselineskip,
  scale=0.75,
  color=blue!70!black,
  no-aka=0
]{pongtex}

\pongtexsetup{height=1.5\baselineskip,scale=0.75,color=teal!65!black}
```

Option	Default	Description
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height	\baselineskip	Height of one upright tile.
scale	0.75	Scale factor for the tile symbol relative to the tile face.
tile-dir	tiles	Directory containing the tile PDF files.
color	none	Recolour the tile back. Use any xcolor colour expression, or none for the embedded artwork.
no-aka	0	Replace red fives 0m, 0p, and 0s with regular fives when set to 1.
river-cols	6	Number of tiles per discard-river row.
river-row-gap	0pt	Vertical gap between discard-river rows.

3 The \mahjong command

```
\mahjong{<tiles>}
\mahjong[<height>][<scale>][<back-colour>][<local-keys>]{<tiles>}
\mahjong[<local-keys>]{<tiles>}
```

The first optional argument also accepts a key-value list. This allows compact local overrides such as:

```
\mahjong[height=1.5\baselineskip,scale=0.75,color=teal!65!black,no-aka=1]{x 0m0p0s}
```

4 MPSZ notation

A tile is written as one or more digits followed by a suit letter. The suits are **m** for manzu, **p** for pinzu, **s** for souzu, and **z** for honors.

Input	Output
111m456s111p11122z	
1112345678999p	
19m19s19p1234567z	

4.1 Special tokens

Token	Meaning
0m, 0p, 0s	Red fives:
x	Face-down tile:
?	Unknown blank tile:
-	Full visual gap between groups.
N-	Proportional gap of N/7 of one tile width, for example 2-.

4.2 Rotated and stacked tiles

The marker ***** or an apostrophe rotates the preceding tile sideways. The marker **+** or a double quote stacks two sideways copies of the preceding tile, which is useful for kan notation.

Input	Output
111m111s111p22z2-3*333z	

123m-xx?4p'5p"0s



4.3 Red fives and the no-aka option

Red fives are enabled by default.

Input	Output
0m0p0s55555m	
\mahjong	
[no-aka=1]{0m0p0s55555p}	

4.4 Concealed kongs

Five identical consecutive digits followed by a suit are rendered as a concealed kan: a face-down tile, two visible tiles, and a face-down tile. For suited fives, the visible pair contains a red five and a regular five unless `no-aka=1` is active.

Input	Output
55555p	
33333z	
\mahjong [no-aka=1]{55555s}	

5 Tile-back recolouring

The tile-back colour can be set globally or locally. The colour value is passed to `xcolor`. The special value `none` restores the embedded back tile.

```
\pongtexsetup{color=purple!70!black}
\mahjong{x x x}
```

```
\mahjong[color=teal!65!black]{x x x}
\mahjong[color=none]{x x x}
```



6 Discard rivers

The `\mahjongriver` command uses the same notation as `\mahjong`, but it breaks tiles into rows. The row length is controlled by `river-cols`.

```
\mahjongriver{1m9m2z5z3s6s1s1m8p4z87m}
\mahjongriver[river-cols=6]{1m9m2z5z3s6s1s*1m8p4z87m}
```



7 Overlay Notes

The `\mahjong` command supports text annotations. Use `[]` in `\mahjong` to attach a note above the previous tile.

Input	Output
<code>111m111s111p2z2-3*333z7-2z[waiting]</code>	
<code>33333z[concealed kong]</code>	

When a note is added to the fifth tile on the concealed kong, a warning `Overlay text '<overlay text>'` is attached to the final back tile of a concealed kong. will be triggered and the note will be moved to the fourth tile.

8 Expl3 interface

For package authors who prefer an expl3-style interface, PongTeX also exposes:

```
\ExplSyntaxOn
\pongtex_typeset_hand:n {111m456s111p11122z}
\pongtexriver:n {111m456s111p11122z}
\ExplSyntaxOff
```

9 Attribution and licence

The LaTeX package code is distributed under the MIT License. Portions are based on Daniel Schmitz's `mahjong` package, copyright 2021 Daniel Schmitz, also under the MIT License.

The tile artwork in `tiles/` is derived from FluffyStuff's `riichi-mahjong-tiles` project. The upstream licence file places that work in the public domain/CC0. Attribution is retained here and in the README.